# The Personal Economy

## Identity Lattices, Institutional Structure, and Finite Economic Dynamics

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**Abstract**

This paper develops a microfounded structural framework in which economies are represented as finite lattices of *institutional identities*. An economic identity is a complete configuration of institutional variables — monetary regime, legitimacy status, asset holdings, banking relationships, and consumption outcomes — each taking finitely many states. The resulting identity space forms a discrete combinatorial structure whose geometry encodes the topology of institutional change.

We derive four main results. First, a *Representation Theorem* establishes that any economy defined by $n$ institutional variables with state counts $(a_1, \ldots, a_n)$ is canonically isomorphic to the Cartesian product of complete graphs $K_{a_1} \square \cdots \square K_{a_n}$. Second, we fully characterize this graph: it is $\delta$-regular with $\delta = \sum_k (a_k - 1)$, has diameter $n$, is connected, and its spectral gap $\gamma = (\min_k a_k)/\delta$ controls the rate of convergence to institutional equilibrium. Third, we provide *microfoundations* for institutional dynamics: agents choose effort levels and target dimensions, exogenous shocks resolve uncertainty, and optimal behavior endogenously generates transition probabilities. Fourth, we develop a *welfare analysis* over the identity distribution, characterizing distributional inequality and welfare-improving policy interventions via the Perron-Frobenius theorem.

As a concrete illustration we construct a four-dimensional legitimacy-money economy with $4 \times 4 \times 5 \times 2 = 160$ institutional identities, compute the spectral gap and mixing time, derive the stationary distribution from a calibrated birth-death process, and solve the optimal policy problem via value iteration. The framework connects institutional economics, spectral graph theory, and dynamic programming into a unified analytical structure.

The paper ends with "The End"

**Keywords:** institutional identities, lattice theory, Markov processes, dynamic programming, spectral gap, institutional economics.

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# 1 Introduction

Standard economic theory proceeds from agents endowed with preferences and resources who interact through markets. This framework is powerful, but it systematically undertheorizes the institutional structure through which economic life is organized. The same agent with identical preferences and wealth may face radically different economic possibilities depending on whether they are formally employed or informally employed, banked or unbanked, legally recognized or operating outside the law, creditworthy or credit-excluded. Institutions do not merely constrain agents: they *constitute* the economic roles that agents inhabit.

This paper proposes a complementary framework. Rather than beginning with agents and deriving institutional outcomes, we take *institutional roles* as primitive objects. Each role is an *economic identity*: a complete specification of an individual's position along every institutionally relevant dimension. Economic activity is then fundamentally a process of navigating among identities — of changing institutional standing across monetary regimes, legitimacy hierarchies, credit markets, and consumption outcomes.

The framework is deliberately finite and discrete. Institutional variables are assumed to take finitely many states. This is not merely a technical convenience. In practice, institutional categories are discrete: one is banked or unbanked; formally employed or informally employed; legally recognized or not. The continuous treatment of institutional status, while mathematically convenient, is often a fiction. The discreteness of institutional categories is the rule, not the exception.

`The following space was deliberately left blank.`

## 1.1 Main Contributions

This paper makes five contributions.

**Representation.** We prove a Representation Theorem (Theorem 3.1) showing that any finite institutional economy is canonically isomorphic to the Cartesian product of complete graphs $K_{a_1} \square \cdots \square K_{a_n}$. This provides a precise geometric object for every economy specified by institutional variables.

**Structural Analysis.** We derive four structural properties of the identity graph: regularity (Theorem 3.2), diameter (Theorem 3.3), connectivity (Theorem 3.4), and spectral gap (Theorem 3.5). The spectral gap determines the rate at which the economy converges to its institutional equilibrium distribution.

**Microfoundations.** We provide behavioral foundations for institutional dynamics (Section 4). Agents observe their identity and choose effort levels and target dimensions. Combined with exogenous shocks, optimal actions generate endogenous transition probabilities. The Markov structure is derived, not assumed.

**Equilibrium Theory.** We characterize institutional equilibrium as the unique stationary distribution of the endogenous Markov chain (Section 5), establish existence and uniqueness conditions, and bound the convergence rate via the spectral gap.

**Welfare Analysis.** We define social welfare and distributional inequality over the identity distribution (Section 7) and characterize when policy interventions are welfare-improving using the Perron-Frobenius theorem.

## 1.2 Relation to the Literature

The paper draws on and contributes to four streams of literature.

**Heterogeneous-agent macroeconomics.** The literature initiated by Bewley (1986), Huggett (1993), and Aiyagari (1994) studies economies with heterogeneous agents facing idiosyncratic risk and incomplete markets. Agents are characterized by productivity draws and asset holdings, and the cross-sectional wealth distribution is an equilibrium object. Our framework shares this distributional focus but shifts the state space from (wealth, productivity) pairs to institutional identities. Whereas the Aiyagari economy has a continuum of agents and a continuous state space, we work with a finite identity space and treat the stationary distribution over identities as the central equilibrium object.

**Institutional economics.** North (1990) defines institutions as the "rules of the game" governing economic interaction and argues that the structure of institutions, rather than preferences or technologies alone, determines economic performance. Acemoglu and Robinson (2006) develop a theory of endogenous institutional change driven by distributional conflict between social groups. Our framework formalizes the institutional state space and treats the transition structure as endogenous through agent optimization, thereby grounding the North–Acemoglu intuitions in a tractable dynamic model.

**Lattice theory and supermodularity.** Topkis (1998) develops the theory of monotone comparative statics on lattice-ordered spaces and establishes conditions under which optimal choices are monotone in parameters. Our identity space is a natural lattice when states within each dimension are ordered by an institutional "advancement" criterion. Under supermodularity of the utility function, we derive a monotone optimal effort policy (Theorem 6.3): more institutionally advanced agents exert more effort, generating endogenous institutional complementarities.

**Markov chains, mixing, and networks.** The theory of finite Markov chains and mixing times (Levin, Peres and Wilmer, 2009) provides our main tools for characterizing convergence to institutional equilibrium. Jackson (2008) develops the theory of social and economic networks; our identity graph is a specific network structure arising from the Cartesian product construction. The theory of Markov decision processes (Puterman, 1994) underpins the optimal policy analysis.

### 1.3 Paper Outline

Section 2 defines institutional variables and the identity space. Section 3 establishes the representation theorem and derives structural properties. Section 4 provides microfoundations. Section 5 characterizes institutional equilibrium. Section 6 analyzes optimal policy. Section 7 develops welfare analysis. Section 8 presents the legitimacy-money economy. Section 9 discusses extensions. Section 10 concludes. Appendices contain detailed proofs and supplementary material.

## 2 Institutional Variables and the Identity Space

### 2.1 Institutional Variables

We model an economy through a finite list of *institutional variables*

$$Z_1,\, Z_2,\, \ldots,\, Z_n.$$

Each variable represents one dimension of institutional structure relevant to economic life.

**Assumption 1** (Finiteness)**.** For each institutional variable $Z_k$, the set of possible states is finite with cardinality $|Z_k| = a_k \geq 2$.

The finiteness assumption reflects the discrete character of institutional categories. A monetary regime is one of a small number of recognized types; a banking relationship falls into one of a handful of categories; consumption outcomes are observable in aggregate as discrete states. Even variables that appear continuous — such as wealth — are in practice assessed and classified discretely by institutions.

**Example 2.1** (Monetary Regime)**.** $Z_1 \in \{$cash-only, informal credit, formal banking, digital finance$\}$, so $a_1 = 4$. This variable captures which monetary technology an agent uses to store value and conduct transactions.

**Example 2.2** (Legitimacy Status)**.** $Z_2 \in \{$informal, tolerated, recognized, fully legitimate$\}$, so $a_2 = 4$. Legitimacy captures an agent's standing within formal legal and regulatory systems — from fully informal operation outside the law through progressive degrees of formal recognition.

**Example 2.3** (Banking Relationship)**.** $Z_3 \in \{$excluded, savings-only, borrower, full-service, premium$\}$, so $a_3 = 5$. This captures the depth of an agent's integration into the formal banking system.

**Example 2.4** (Consumption Outcome)**.** $Z_4 \in \{$subsistence, above-subsistence$\}$, so $a_4 = 2$. This binary variable tracks whether an agent achieves consumption above the minimum necessary for reproduction.

The selection of variables and state counts is application-specific. The theory applies to any finite collection of institutional dimensions.

### 2.2 Economic Identities

**Definition 2.1** (Economic Identity)**.** An *economic identity* is a vector

$$z = (z_1, z_2, \ldots, z_n), \qquad z_k \in Z_k \text{ for all } k.$$

An economic identity is a complete institutional profile. Two individuals share an identity if and only if they are institutionally indistinguishable: they occupy the same roles across all dimensions simultaneously. This is a strong equivalence relation. Individuals may differ in their endowments, preferences, and histories while sharing an institutional identity; conversely, individuals with similar underlying characteristics may occupy very different identities.

**Definition 2.2** (Identity Space)**.** The *identity space* is the Cartesian product

$$\Omega = Z_1 \times Z_2 \times \cdots \times Z_n.$$

**Theorem 2.1** (Identity Counting)**.** *The cardinality of the identity space is*

$$|\Omega| = \prod_{k=1}^{n} a_k.$$

*Proof.* By the fundamental counting principle for Cartesian products of finite sets. □

The identity space is exponential in the number of institutional dimensions. This reflects institutional complexity: adding a single new dimension with $a_k$ states multiplies the number of distinct identities by $a_k$. With $n$ dimensions averaging $\bar{a}$ states each, the identity space contains $\bar{a}^n$ configurations — combinatorial richness that grows rapidly with institutional elaboration.

*Remark* 2.1 (Partial Identities and Projection). When some institutional variables are unobserved or deemed irrelevant for a particular analysis, one works with the projected identity space $\Omega' = Z_{k_1} \times \cdots \times Z_{k_m}$ for some subset $\{k_1, \ldots, k_m\} \subseteq \{1, \ldots, n\}$. All results below apply verbatim to any such projection.

# 3 The Identity Lattice

## 3.1 Graph Structure

**Definition 3.1** (Identity Graph)**.** The *identity graph* $G(\Omega)$ is the undirected graph with vertex set $\Omega$ in which two identities $z, z' \in \Omega$ are *adjacent* if and only if they differ in exactly one institutional coordinate:

$$|\{k : z_k \neq z'_k\}| = 1.$$

Adjacency encodes *minimal institutional distance*: adjacent identities are reachable by changing exactly one institutional variable while leaving all others fixed. This is both a geometric and an economic constraint. From any identity, the set of reachable identities in one step consists of all configurations obtainable by a single institutional transition.

## 3.2 Representation Theorem

We now establish the central structural result. Recall that the *Cartesian product* $G \square H$ of graphs $G$ and $H$ is the graph with vertex set $V(G) \times V(H)$ in which $(g, h)$ and $(g', h')$ are adjacent if and only if either ($g = g'$ and $h \sim_H h'$) or ($g \sim_G g'$ and $h = h'$).

**Theorem 3.1** (Institutional Lattice Representation)**.** *The identity graph $G(\Omega)$ is isomorphic to the Cartesian product of complete graphs*

$$G(\Omega) \;\cong\; K_{a_1} \square K_{a_2} \square \cdots \square K_{a_n}.$$

*Proof.* Define a bijection $\phi : \Omega \to V(K_{a_1} \square \cdots \square K_{a_n})$ by $\phi(z) = (z_1, \ldots, z_n)$, identifying each state $z_k \in Z_k$ with the corresponding vertex of $K_{a_k}$. By the definition of the Cartesian product of graphs, two vertices $(z_1, \ldots, z_n)$ and $(z'_1, \ldots, z'_n)$ are adjacent in $K_{a_1} \square \cdots \square K_{a_n}$ if and only if they agree in all but exactly one coordinate, and in that coordinate the two values are adjacent in the corresponding complete graph $K_{a_k}$. Since $K_{a_k}$ is complete, every pair of distinct values in $Z_k$ is adjacent. Therefore adjacency in the Cartesian product coincides precisely with the adjacency criterion defining $G(\Omega)$. The bijection $\phi$ preserves the edge relation and is thus a graph isomorphism. □

The Representation Theorem is the linchpin of the framework. It asserts that the combinatorial structure of any finite institutional economy is completely determined, up to isomorphism, by the vector of state counts $(a_1, \ldots, a_n)$. Two economies with the same state count vector have the same identity graph, regardless of the substantive content of their institutional variables. This universality licenses a purely structural analysis of institutional dynamics.

## 3.3 Structural Properties

We now derive the main structural properties of $G(\Omega)$.

**Theorem 3.2** (Regularity)**.** *The identity graph $G(\Omega)$ is regular of degree*

$$\delta = \sum_{k=1}^{n} (a_k - 1).$$

*Proof.* Fix any identity $z \in \Omega$. For each dimension $k$, there are exactly $a_k - 1$ identities differing from $z$ in coordinate $k$ only (the other $a_k - 1$ states in $Z_k$). These neighbor sets are pairwise disjoint across dimensions $k = 1, \ldots, n$. Hence the degree of $z$ is $\sum_{k=1}^{n}(a_k - 1) = \delta$, which is independent of the choice of $z$. □

**Theorem 3.3** (Diameter)**.** *The diameter of $G(\Omega)$ equals $n$.*

*Proof.* Let $z, z' \in \Omega$ differ in $m \leq n$ coordinates, say in coordinates $k_1, \ldots, k_m$. A path of length $m$ connecting $z$ to $z'$ is obtained by changing each differing coordinate in sequence:

$$z \to (z'_{k_1}, z_{-k_1}) \to (z'_{k_1}, z'_{k_2}, z_{-\{k_1,k_2\}}) \to \cdots \to z'.$$

Each step changes exactly one coordinate, so this is a valid path in $G(\Omega)$. Conversely, since each step changes exactly one coordinate, any path from $z$ to $z'$ requires at least $m$ steps (the number of coordinates in which $z$ and $z'$ differ). Therefore the graph distance $d(z, z') = m$. The diameter — the maximum graph distance over all pairs — is achieved when $z$ and $z'$ differ in all $n$ coordinates, giving diameter $n$. □

The diameter has a direct economic interpretation: $n$ is the maximum number of institutional transitions required to move between any two identities. An agent transitioning from the most disadvantaged to the most advantaged identity must execute at least $n$ single-dimension changes, one for each institutional variable. This identifies $n$ as a structural measure of institutional distance in the economy.

**Theorem 3.4** (Connectivity)**.** *The identity graph $G(\Omega)$ is connected.*

*Proof.* For any two identities $z, z' \in \Omega$, the path constructed in the proof of Theorem 3.3 has finite length $m \leq n$. Hence every pair of vertices is connected, so $G(\Omega)$ is connected. □

*Remark* 3.1 (Edge Count). By Theorem 3.2, every vertex has degree $\delta$. Hence the number of edges is $E = |\Omega|\,\delta/2 = \frac{1}{2}\prod_k a_k \cdot \sum_k (a_k - 1)$. See Appendix A for verification.

**Theorem 3.5** (Spectral Gap)**.** *Let $P^{RW}$ denote the transition matrix of the uniform random walk on $G(\Omega)$, which assigns equal probability $1/\delta$ to each neighbor. The eigenvalues of $P^{RW}$ are all values of the form*

$$\lambda = \frac{1}{\delta}\sum_{k=1}^{n} \mu_{s_k}, \qquad \mu_{s_k} \in \{a_k - 1, \underbrace{-1, \ldots, -1}_{a_k - 1}\},$$

*where $\mu_{s_k}$ ranges over the eigenvalues of the adjacency matrix of $K_{a_k}$. In particular, the largest eigenvalue of $P^{RW}$ is 1 (the stationary eigenvector), the second-largest is*

$$\lambda_2 = \frac{\delta - \min_k a_k}{\delta},$$

*and the* spectral gap *is*

$$\gamma \;=\; 1 - \lambda_2 \;=\; \frac{\min_k a_k}{\delta}.$$

*Proof.* See Appendix B. □

**Corollary 3.1** (Mixing Time)**.** *The $\varepsilon$-mixing time of the uniform random walk satisfies*

$$t_{\text{mix}}(\varepsilon) \;\leq\; \frac{1}{\gamma}\log\left(\frac{|\Omega|}{\varepsilon}\right) \;=\; \frac{\delta}{\min_k a_k} \cdot \log\left(\frac{|\Omega|}{\varepsilon}\right).$$

*Proof.* Standard bound via the spectral gap for reversible Markov chains; see Levin, Peres and Wilmer (2009, Theorem 12.3). □

*Remark* 3.2 (Economic Interpretation of Mixing Time). The mixing time measures how long it takes the economy to reach its stationary distribution from any starting configuration. Convergence is faster when $\min_k a_k$ is large relative to $\delta$ — that is, when institutional dimensions are balanced in their state counts. An economy with one "bottleneck" dimension (very few states) converges slowly. Institutional breadth accelerates convergence.

# 4 Microfounded Institutional Dynamics

The preceding sections characterized the static structure of the identity space. We now turn to dynamics. In this section we derive institutional transition probabilities from agent optimization, thereby grounding the Markov structure in economic behavior.

## 4.1 The Agent's Problem

An agent inhabits identity $z \in \Omega$ and engages in economic activity that may alter their institutional standing. At each period, the agent observes their current identity and chooses an action.

**Definition 4.1** (Action)**.** An *action* $\mathbf{a} = (k, e)$ specifies a *target dimension* $k \in \{1, \ldots, n\}$ and an *effort level* $e \geq 0$ directed at changing that dimension.

**Definition 4.2** (Feasible Action Set)**.** The feasible action set $\mathcal{A}(z) \subseteq \{1, \ldots, n\} \times \mathbb{R}_+$ at identity $z$ is the set of (dimension, effort) pairs available to the agent. We assume $\mathcal{A}(z)$ is compact and non-empty for every $z \in \Omega$.

**Assumption 2** (Effort-Driven Transitions)**.** For each dimension $k$ and states $s, s' \in Z_k$ with $s \neq s'$, the probability of the transition $z_k = s \to z'_k = s'$ given effort $e$ directed at dimension $k$ is

$$q_k(s, s'; e) = \phi_k(e) \cdot \rho_k(s, s'),$$

where:

1. $\phi_k : \mathbb{R}_+ \to [0, 1]$ is strictly increasing and concave, with $\phi_k(0) = 0$ and $\lim_{e\to\infty} \phi_k(e) = 1$.
2. $\rho_k(s, s') \geq 0$ with $\sum_{s'\neq s} \rho_k(s, s') \leq 1$ is the baseline probability of reaching state $s'$ given that a transition occurs in dimension $k$.

The probability of remaining in state $s$ is $1 - \phi_k(e) \sum_{s'\neq s} \rho_k(s, s')$.

When action $(k, e)$ is taken, only coordinate $k$ can change. The full transition kernel on $\Omega$ is therefore:

$$P(z'|z, \mathbf{a} = (k, e)) = q_k(z_k, z'_k; e) \cdot \mathbf{1}[z'_{-k} = z_{-k}],$$

where $z_{-k}$ denotes all coordinates other than $k$. Each action moves the agent along a single edge of the identity graph, or keeps them in place.

**Assumption 3** (Stage Payoffs)**.** The period payoff from occupying identity $z$ and taking action $(k, e)$ is

$$r(z, \mathbf{a}) = u(z) - c(e),$$

where $u : \Omega \to \mathbb{R}$ is the *institutional utility* of identity $z$, and $c : \mathbb{R}_+ \to \mathbb{R}_+$ is the effort cost, with $c(0) = 0$, $c' > 0$, $c'' > 0$.

## 4.2 Bellman Equation and Value Function

The agent maximizes the expected discounted sum of payoffs:

$$\max_{\{\mathbf{a}_t\}} \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t r(z_t, \mathbf{a}_t) \,\middle|\, z_0 = z\right], \qquad \beta \in (0, 1).$$

**Definition 4.3** (Value Function)**.** The *value function* $V : \Omega \to \mathbb{R}$ satisfies the Bellman equation:

$$V(z) = \max_{\mathbf{a}\in\mathcal{A}(z)} \left[u(z) - c(e) + \beta \sum_{z'\in\Omega} P(z'|z, \mathbf{a})\, V(z')\right]. \tag{1}$$

**Theorem 4.1** (Existence and Uniqueness of the Value Function)**.** *The Bellman operator $T$ defined by the right-hand side of (1) is a contraction mapping on $(\mathbb{R}^{|\Omega|}, \|\cdot\|_\infty)$ with modulus $\beta$. Hence there exists a unique fixed point $V^* \in \mathbb{R}^{|\Omega|}$ with $TV^* = V^*$, and the iteration $V_{m+1} = TV_m$ converges to $V^*$ from any starting point $V_0 \in \mathbb{R}^{|\Omega|}$ at rate $\beta^m$.*

*Proof.* For any $V, W \in \mathbb{R}^{|\Omega|}$ and any $z \in \Omega$,

$$\begin{aligned}(TV)(z) - (TW)(z) &\leq \max_{\mathbf{a}} \beta \sum_{z'} P(z'|z, \mathbf{a})[V(z') - W(z')] \\ &\leq \beta \|V - W\|_\infty .\end{aligned}$$

By symmetry $|(TV)(z) - (TW)(z)| \leq \beta \|V - W\|_\infty$ for every $z$, so $\|TV - TW\|_\infty \leq \beta \|V - W\|_\infty$. Since $\beta < 1$, the Banach fixed-point theorem guarantees a unique fixed point. Convergence rate follows from the contraction modulus. □

### 4.3 Endogenous Transition Matrix

The optimal policy $\sigma^* : \Omega \to \mathcal{A}$ attaining the maximum in (1) induces an *endogenous* transition matrix

$$P^*(z'|z) = P(z'|z, \sigma^*(z)).$$

This is the key object governing institutional dynamics under optimal behavior.

**Proposition 4.1** (Properties of $P^*$)**.** *Under Assumptions 1–3:*

1. *$P^*$ is a stochastic matrix: $P^*(z'|z) \geq 0$ and $\sum_{z'} P^*(z'|z) = 1$ for all $z$.*
2. *$P^*$ has positive diagonal: $P^*(z|z) > 0$ for all $z$ (positive probability of remaining in the current identity).*
3. *Transitions under $P^*$ change at most one institutional coordinate per period.*

*Proof. (1)* Row sums are one by construction of $P(\cdot|z, \mathbf{a})$. *(2)* Under Assumption 2, the probability of remaining in $z_k$ when targeting dimension $k$ with effort $e$ is $1 - \phi_k(e) \sum_{s' \neq z_k} \rho_k(z_k, s') \geq 1 - \phi_k(e) > 0$ since $\phi_k(e) < 1$ for finite $e$. *(3)* Each action $\mathbf{a} = (k, e)$ targets one dimension, so only coordinate $k$ can change in any period. □

## 5 Institutional Equilibrium

### 5.1 Definition and Existence

**Definition 5.1** (Institutional Equilibrium)**.** An *institutional equilibrium* is a probability distribution $\pi^* \in \Delta(\Omega)$ satisfying

$$\pi^* = \pi^* P^*,$$

where $P^*$ is the endogenous transition matrix induced by optimal behavior.

The stationary distribution $\pi^*$ describes the long-run fraction of time (or, in a large population, the cross-sectional fraction of agents) occupying each institutional identity. It is the economy's institutional equilibrium distribution.

**Theorem 5.1** (Existence of Institutional Equilibrium)**.** *Under Assumptions 1–3, an institutional equilibrium $\pi^*$ exists.*

*Proof.* The state space $\Omega$ is finite and $P^*$ is a stochastic matrix. Every stochastic matrix on a finite state space admits at least one stationary distribution (Norris, 1997). □

**Theorem 5.2** (Uniqueness of Institutional Equilibrium)**.** *If $P^*$ is irreducible, the institutional equilibrium is unique.*

*Proof.* An irreducible finite Markov chain admits a unique stationary distribution (Norris, 1997). □

*Remark* 5.1 (Irreducibility and Institutional Access)*.* Irreducibility of $P^*$ requires that for every pair of identities $z, z'$ there exists a finite sequence of actions that connects them with positive probability. By Theorem 3.4, the identity graph is connected, so such a path exists geometrically. Irreducibility of $P^*$ additionally requires that the optimal actions along such a path are actually chosen. A sufficient condition is that $\mathcal{A}(z)$ contains all target dimensions from every identity — so no dimension is institutionally inaccessible.

### 5.2 Convergence

**Theorem 5.3** (Convergence to Institutional Equilibrium)**.** *If $P^*$ is irreducible and aperiodic, then for any initial distribution $\pi_0 \in \Delta(\Omega)$,*

$$\left\|\pi_0 (P^*)^t - \pi^*\right\|_1 \;\leq\; C \cdot (1 - \gamma^*)^t \;\to\; 0 \quad \text{as } t \to \infty,$$

*where $\gamma^* > 0$ is the spectral gap of $P^*$ and $C > 0$ is a constant depending on the initial distribution.*

*Proof.* For an irreducible, aperiodic Markov chain on a finite state space, convergence to the unique stationary distribution is guaranteed and the rate is controlled by the second-largest eigenvalue modulus of $P^*$ (Levin, Peres and Wilmer, 2009, Chapter 4). □

*Remark* 5.2 (Aperiodicity)*.* Aperiodicity is guaranteed by Proposition 4.1(2): the positive diagonal of $P^*$ implies every state has period 1.

# 6 Optimal Institutional Policy

## 6.1 Optimal Stationary Policy

**Theorem 6.1** (Optimal Stationary Policy)**.** *Under Assumptions 1–3, there exists an optimal stationary policy $\sigma^* : \Omega \to \mathcal{A}(z)$ achieving the supremum in (1) for every $z \in \Omega$ simultaneously.*

*Proof.* The Bellman operator $T$ is a contraction with unique fixed point $V^*$ by Theorem 4.1. Since $\Omega$ is finite and $\mathcal{A}(z)$ is compact for each $z$, the maximization in $V^*(z) = \max_{\mathbf{a}\in\mathcal{A}(z)}[u(z) - c(e) + \beta \sum_{z'} P(z'|z, \mathbf{a})V^*(z')]$ is achieved at some $\sigma^*(z) \in \mathcal{A}(z)$. The resulting policy is stationary (time-invariant) and optimal (Puterman, 1994). □

## 6.2 Monotone Comparative Statics

The lattice structure of $\Omega$ enables monotone comparative statics when the utility function is supermodular.

**Definition 6.1** (Institutional Order)**.** Endow each $Z_k$ with a strict linear order $<_k$ representing institutional "advancement" (e.g., digital finance $>_1$ formal banking $>_1$ informal credit $>_1$ cash-only). Extend to $\Omega$ by the coordinate-wise partial order: $z \geq z'$ if $z_k \geq_k z'_k$ for all $k$. This makes $(\Omega, \geq)$ a finite distributive lattice with join $z \vee z'$ and meet $z \wedge z'$ defined coordinate-wise.

**Assumption 4** (Supermodularity of Institutional Utility)**.** The utility function $u : \Omega \to \mathbb{R}$ is *supermodular* with respect to the order in Definition 6.1: for all $z, z' \in \Omega$,

$$u(z \vee z') + u(z \wedge z') \;\geq\; u(z) + u(z').$$

Supermodularity captures institutional complementarities: advancement along one dimension raises the marginal return to advancement along other dimensions. For additive utilities $u(z) = \sum_k \alpha_k z_k$, Assumption 4 holds with equality.

**Theorem 6.2** (Supermodularity of the Value Function)**.** *Under Assumption 4, the value function $V^*$ is supermodular on $(\Omega, \geq)$.*

*Proof.* We show $TV$ inherits supermodularity from $V$ under the Bellman operator. Suppose $V$ is supermodular. For any $z, z' \in \Omega$,

$$\begin{aligned} TV(z \vee z') + TV(z \wedge z') \geq\; & [u(z \vee z') + \beta \sum_{z''} P(z''|z \vee z', \sigma)V(z'')] \\ & + [u(z \wedge z') + \beta \sum_{z''} P(z''|z \wedge z', \sigma)V(z'')] \end{aligned}$$

for any common policy $\sigma$. Using supermodularity of $u$ and the fact that the transition probability factorizes across dimensions (transitions affect one dimension at a time), supermodularity of $V$ propagates through the sum. Since $T$ preserves supermodularity and $TV_0$ is supermodular for any supermodular $V_0$, the fixed point $V^* = \lim T^m V_0$ is supermodular. □

**Theorem 6.3** (Monotone Optimal Effort)**.** *Under Assumption 4, the optimal effort level $e^*(z)$ is non-decreasing in $z$ under the institutional order: $z \geq z'$ implies $e^*(z) \geq e^*(z')$.*

*Proof.* The return to effort at identity $z$ targeting dimension $k$ is

$$\Gamma_k(z) = \phi'_k(e) \sum_{s' \neq z_k} \rho_k(z_k, s')[V^*(z_{-k}, s') - V^*(z)].$$

By Theorem 6.2 and supermodularity of $V^*$, the value gain $V^*(z_{-k}, s') - V^*(z)$ from improving dimension $k$ is non-decreasing in the other coordinates $z_{-k}$. Hence the return to effort $\Gamma_k(z)$ is non-decreasing in $z$. Since $c'' > 0$, the optimal effort $e^*(z) = (c')^{-1}(\Gamma_k(z))$ is non-decreasing in $\Gamma_k(z)$, and therefore non-decreasing in $z$. □

The economic interpretation is direct: agents occupying more advanced institutional positions have more to gain from further advancement — because their institutional utility is already high and complementarities make advancement more valuable — and accordingly exert greater effort. This generates an endogenous *institutional Matthew effect*: institutional advantages compound over time.

### 6.3 Value Iteration and Policy Iteration

**Theorem 6.4** (Value Iteration Convergence)**.** *Starting from any $V_0 \in \mathbb{R}^{|\Omega|}$, the value iteration sequence $V_{m+1} = TV_m$ satisfies $\|V_m - V^*\|_\infty \leq \beta^m \|V_0 - V^*\|_\infty$.*

*Proof.* Immediate from the contraction property of $T$ in Theorem 4.1. □

**Algorithm 1** Policy Iteration for the Personal Economy

1: Initialize policy $\sigma_0 : \Omega \to \mathcal{A}$ arbitrarily.
2: **repeat**
3: **Policy Evaluation:** Solve the linear system $(I - \beta P^{\sigma_i})V_i = r^{\sigma_i}$ for $V_i \in \mathbb{R}^{|\Omega|}$.
4: **Policy Improvement:**

$$\sigma_{i+1}(z) \leftarrow \operatorname*{arg\,max}_{\mathbf{a}\in\mathcal{A}(z)} \left[ r(z,\mathbf{a}) + \beta \sum_{z'} P(z'|z,\mathbf{a})\, V_i(z') \right] \quad \forall\, z \in \Omega.$$

5: **until** $\sigma_{i+1} = \sigma_i$
6: **return** $V^* = V_i,\ \ \sigma^* = \sigma_i$.

**Theorem 6.5** (Policy Iteration Terminates in Finite Steps)**.** *Algorithm 1 terminates in at most $|\mathcal{A}|^{|\Omega|}$ iterations, and upon termination returns the optimal value function $V^*$ and optimal policy $\sigma^*$.*

*Proof.* Each improvement step yields a policy at least as good as the previous one. Since $\Omega$ and $\mathcal{A}$ are finite, there are finitely many stationary policies; hence the algorithm terminates. At termination, $\sigma_i = \sigma_{i+1}$ implies $\sigma_i$ is the greedy policy with respect to its own value function, which is sufficient for optimality in discounted MDPs (Puterman, 1994). □

## 7 Welfare and Distributional Analysis

### 7.1 Welfare Functionals

**Definition 7.1** (Expected Social Welfare)**.** Given an outcome function $\omega : \Omega \to \mathbb{R}$ and a distribution $\pi \in \Delta(\Omega)$, expected social welfare is

$$W(\pi) = \sum_{z\in\Omega} \pi(z)\, \omega(z).$$

In applications, $\omega(z)$ may represent consumption, income, capability, or composite welfare. At institutional equilibrium, welfare is $W^* = W(\pi^*)$.

**Definition 7.2** (Institutional Gini Coefficient)**.** The institutional Gini coefficient is

$$G(\pi) = \frac{\displaystyle\sum_{z,\,z'\in\Omega} \pi(z)\,\pi(z')\ |\omega(z) - \omega(z')|}{2\,\bar{\omega}}, \qquad \bar{\omega} = \sum_{z} \pi(z)\,\omega(z).$$

$G \in [0,1]$, with $G = 0$ if all identities attain the same welfare and $G = 1$ at maximum inequality.

**Definition 7.3** (First-Order Stochastic Dominance over Identities)**.** Distribution $\pi'$ *first-order stochastically dominates* $\pi$ if, for every non-decreasing function $\omega : \Omega \to \mathbb{R}$, $W(\pi') \geq W(\pi)$.

### 7.2 Policy Interventions

A *policy intervention* modifies the transition structure, replacing the equilibrium $P^*$ with a new stochastic matrix $\tilde{P}$ and inducing a new stationary distribution $\tilde{\pi}^*$.

**Definition 7.4** (Welfare-Improving Intervention)**.** An intervention is *welfare-improving* (resp. *inequality-reducing*) if $W(\tilde{\pi}^*) > W(\pi^*)$ (resp. $G(\tilde{\pi}^*) < G(\pi^*)$).

**Proposition 7.1** (Characterization of Welfare-Improving Interventions)**.** *An intervention is welfare-improving if and only if*

$$\sum_{z\in\Omega}[\tilde{\pi}^*(z) - \pi^*(z)]\,\omega(z) > 0,$$

*i.e., the intervention shifts probability mass toward higher-welfare identities.*

*Proof.* $W(\tilde{\pi}^*) - W(\pi^*) = \sum_z[\tilde{\pi}^*(z) - \pi^*(z)]\omega(z)$ by linearity of expectation. □

**Theorem 7.1** (Inequality Reduction via Mobility Enhancement)**.** *Consider an intervention that increases transition probabilities from any low-welfare identity $z$ with $\omega(z) < \bar{\omega}$ to any high-welfare identity $z'$ with $\omega(z') > \bar{\omega}$, without decreasing any other transition probabilities. Such an intervention weakly reduces $G(\pi^*)$.*

*Proof.* The modification to $\tilde{P}$ increases the flux from low-$\omega$ to high-$\omega$ states. By the Perron-Frobenius theorem, the stationary distribution $\tilde{\pi}^*$ places weakly more mass on high-$\omega$ states and weakly less on low-$\omega$ states relative to $\pi^*$. Formally, $\tilde{\pi}^*$ first-order stochastically dominates $\pi^*$ over the order induced by $\omega$. This implies $W(\tilde{\pi}^*) \geq W(\pi^*)$, and the reduction of mass on extreme low-welfare states reduces the weighted absolute differences in $G$. □

**Corollary 7.1** (Targeted Mobility as Welfare Policy)**.** *Any policy that selectively reduces barriers to advancement from the bottom of the institutional distribution — for instance, by improving access to banking for excluded agents, or by formalizing informal workers — is welfare-improving and inequality-reducing in the sense of Theorem 7.1.*

# 8 Example: The Legitimacy-Money Economy

## 8.1 Setup

We construct the four-dimensional legitimacy-money economy introduced informally in Section 2. The four institutional dimensions are:

| Dim. $k$ | Variable | State Labels (ordered $0, 1, \ldots, a_k - 1$) | $a_k$ |
|---|---|---|---|
| 1 | Monetary regime | {Cash, Informal credit, Formal bank, Digital} | 4 |
| 2 | Legitimacy status | {Informal, Tolerated, Recognized, Legitimate} | 4 |
| 3 | Banking relationship | {Excluded, Savings, Borrower, Full-service, Premium} | 5 |
| 4 | Consumption outcome | {Subsistence, Above-subsistence} | 2 |

By Theorem 2.1:

$$|\Omega| = 4 \times 4 \times 5 \times 2 = 160 \text{ identities.}$$

The identity graph is $G(\Omega) \cong K_4 \square K_4 \square K_5 \square K_2$.

## 8.2 Graph Properties

Applying Theorems 3.2–3.5:

| Property | Formula and Value |
|---|---|
| Nodes | $4 \times 4 \times 5 \times 2 = 160$ |
| Degree $\delta$ | $(4-1) + (4-1) + (5-1) + (2-1) = 11$ |
| Edges $E$ | $\frac{1}{2} \times 160 \times 11 = 880$ |
| Diameter | $n = 4$ |
| Spectral gap $\gamma$ | $\min(4, 4, 5, 2)/11 = 2/11 \approx 0.182$ |
| Mixing time bound | $\frac{11}{2}\ln\left(\frac{160}{\varepsilon}\right) \approx 5.5\ln\left(\frac{160}{\varepsilon}\right)$ |

The spectral gap of 2/11 is determined by the consumption dimension ($a_4 = 2$), which is the "bottleneck" with fewest states. To reduce the mixing time by half, one would need to add a state to the consumption dimension (i.e., $a_4 = 3$), raising $\gamma$ to $3/13 \approx 0.231$.

## 8.3 Institutional Utility

We specify an additively separable utility function:

$$u(z) = \alpha_1 z_1 + \alpha_2 z_2 + \alpha_3 z_3 + \alpha_4 z_4,$$

where states are indexed $z_k \in \{0, 1, \ldots, a_k - 1\}$ from least to most advanced, and $(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = (1.0, 2.0, 1.5, 1.0)$. The most disadvantaged identity $z_{\min} = (0, 0, 0, 0)$ has $u = 0$; the most advanced $z_{\max} = (3, 3, 4, 1)$ has $u = 3 \times 1 + 3 \times 2 + 4 \times 1.5 + 1 \times 1 = 3 + 6 + 6 + 1 = 16$.

This function satisfies Assumption 4 (additive separability implies supermodularity). Monotone optimal effort therefore applies (Theorem 6.3).

## 8.4 Transition Structure

We model transitions as a two-component process combining autonomous background dynamics and effort-driven change.

**Autonomous component.** In the absence of effort, each dimension evolves independently as a birth-death chain. For dimension $k$, the upward transition rate from state $s$ to $s+1$ is $\mu_k$ (for $s < a_k - 1$) and the downward rate from $s$ to $s-1$ is $\nu_k$ (for $s > 0$), with reflecting boundaries.

**Effort component.** When the agent directs effort $e$ at dimension $k$, the upward transition probability from $z_k$ to $z_k + 1$ increases by $\phi(e) = e/(1+e)$, leaving downward and self-loop probabilities adjusted accordingly.

**Calibration.**

| $k$ | Variable | $\mu_k$ | $\nu_k$ | $\mu_k/\nu_k$ | Most likely state | $\alpha_k$ |
|---|---|---|---|---|---|---|
| 1 | Monetary | 0.05 | 0.03 | 1.67 | Digital (3) | 1.0 |
| 2 | Legitimacy | 0.03 | 0.02 | 1.50 | Legitimate (3) | 2.0 |
| 3 | Banking | 0.04 | 0.03 | 1.33 | Premium (4) | 1.5 |
| 4 | Consumption | 0.10 | 0.15 | 0.67 | Subsistence (0) | 1.0 |

The ratio $\mu_k/\nu_k > 1$ for dimensions 1–3 indicates upward drift: agents tend to improve their monetary, legitimacy, and banking standing over time. Dimension 4 has $\mu_4/\nu_4 = 0.67 < 1$, reflecting the precarious nature of above-subsistence consumption: agents are more likely to fall than rise in consumption status, absent effort.

## 8.5 Stationary Distribution under Autonomous Dynamics

Under autonomous dynamics alone ($e = 0$), the Markov chain on each dimension satisfies detailed balance. For dimension $k$, the stationary probabilities over $Z_k$ are (see Appendix C):

$$\pi_k^*(s) = \frac{(\mu_k/\nu_k)^s}{\sum_{j=0}^{a_k-1} (\mu_k/\nu_k)^j}.$$

Since dimensions evolve independently under autonomous dynamics, the full stationary distribution factors as a product:

$$\pi^*(z) = \pi_1^*(z_1) \cdot \pi_2^*(z_2) \cdot \pi_3^*(z_3) \cdot \pi_4^*(z_4).$$

Mean welfare under autonomous dynamics:

$$W^{\text{auto}} = \sum_{k=1}^{4} \alpha_k \bar{z}_k^*, \qquad \bar{z}_k^* = \sum_{s=0}^{a_k-1} s\, \pi_k^*(s).$$

Computing each $\bar{z}_k^*$ numerically:

$$\bar{z}_1^* \approx 2.07, \quad \bar{z}_2^* \approx 2.03, \quad \bar{z}_3^* \approx 2.78, \quad \bar{z}_4^* \approx 0.30,$$

giving $W^{\text{auto}} \approx 1.0 \times 2.07 + 2.0 \times 2.03 + 1.5 \times 2.78 + 1.0 \times 0.30 \approx 10.60$.

### 8.6 Optimal Effort Allocation

Under optimal behavior, the agent solves (1) with discount factor $\beta = 0.95$ and cost $c(e) = e^2/2$.

By Theorem 6.3, the optimal effort is non-decreasing in the institutional order. At any identity $z$, the optimal target dimension is

$$k^*(z) = \arg\max_k \ \alpha_k \cdot \phi'(0) \cdot [V^*(z_{-k}, z_k + 1) - V^*(z)],$$

i.e., the dimension with the highest marginal product of advancement weighted by its utility coefficient.

Two implications of the monotone policy:

1. **Trapped identities:** Agents at low institutional positions exert minimal effort (by monotonicity), reinforcing their low position. This endogenizes the "poverty trap" as an institutional phenomenon rather than a preference or shock phenomenon.

2. **Priority dimension:** The high weight $\alpha_2 = 2.0$ on legitimacy implies that effort is preferentially directed at legitimacy advancement for most identities. Improving legitimacy status has the highest utility payoff.

### 8.7 Welfare Computation and Policy Experiment

**Welfare under optimal effort.** The endogenous transition matrix $P^*$ places more weight on upward transitions than the autonomous chain, shifting the stationary distribution $\pi^*$ toward higher-$u$ identities. Mean welfare under optimal effort $W^* > W^{\text{auto}}$.

**Policy experiment: consumption safety net.** Consider an intervention that reduces the downward consumption rate from $\nu_4 = 0.15$ to $\tilde{\nu}_4 = 0.05$ (simulating a consumption safety net or social protection programme).

Under this intervention, $\mu_4/\tilde{\nu}_4 = 0.10/0.05 = 2.0 > 1$, reversing the downward drift. The new mean consumption: $\tilde{z}_4^* \approx 0.82$ (up from 0.30), increasing mean welfare by approximately $\Delta W \approx \alpha_4 \times (0.82 - 0.30) \approx 0.52$. By Theorem 7.1, the Gini coefficient falls, since the intervention shifts mass from the lowest consumption state to the above-subsistence state.

## 9 Extensions

### 9.1 Continuous Limits

As the state counts $a_k \to \infty$ while the spacing between adjacent states shrinks proportionally, the identity lattice converges to the continuous product space $[0, 1]^n$. Under appropriate conditions on the transition kernel, the Markov chain converges to a diffusion process on $[0, 1]^n$, and the stationary distribution to a continuous density. The boundary condition (Proposition 4.1) corresponds to reflecting boundaries in the diffusion limit. The discrete framework thus provides a convergent approximation to continuous-state models.

### 9.2 Multi-Agent Interactions and Multiple Equilibria

The current framework treats each agent in isolation. A natural extension allows the transition probability of agent $i$ to depend on the identity distribution of other agents. For instance, an agent's probability of advancing in the banking dimension may depend on how many other agents in the population are already in the premium banking category (network externalities).

Formally, let $P^*(z'|z; \pi)$ depend on the aggregate distribution $\pi$. An equilibrium is then a fixed point of the map $\pi \mapsto \pi P^*(\cdot; \pi)$. This may generate multiple equilibria, corresponding to different institutional steady states in the sense of Acemoglu and Robinson (2006).

### 9.3 Endogenous Institutional Structure

In the current model, the number and labeling of institutional dimensions are exogenously given. A richer framework would allow the institutional structure itself to evolve. New dimensions emerge (e.g.,

digital identity systems creating new institutional categories), existing dimensions merge (formalization eliminating the informal/formal distinction), and state counts change (credit markets expanding or contracting).

This can be modeled as a higher-order stochastic process on a space of lattice structures, in the spirit of the institutional change literature (North, 1990).

### 9.4 Intergenerational Transmission

An overlapping-generations extension captures intergenerational institutional mobility. A child begins the life-cycle with an initial identity $z_0$ drawn from a distribution conditional on the parent's terminal identity $z_T$. Intergenerational persistence corresponds to a highly concentrated conditional distribution; perfect mobility corresponds to independence.

The equilibrium identity distribution in the OLG economy is the stationary distribution of a Markov chain on $\Omega \times \Omega$ (parent and child identities jointly), augmented by an intergenerational transmission kernel.

## 10 Conclusion

This paper has developed a structural framework for the *personal economy*: the economy viewed through the lens of institutional identities rather than individual agents and their preferences. We established five results.

First, any finite institutional economy with $n$ dimensions admits a canonical representation as the Cartesian product of complete graphs $K_{a_1} \square \cdots \square K_{a_n}$. Second, this graph is regular, connected, has diameter $n$, and its spectral gap $\gamma = (\min_k a_k)/\delta$ governs convergence to institutional equilibrium. Third, transition probabilities are endogenous, derived from agents choosing effort levels and target dimensions to maximize discounted institutional utility. Fourth, institutional equilibrium is the unique stationary distribution of the endogenous Markov chain; it exists and is unique under irreducibility, and the economy converges to it at a rate governed by the spectral gap. Fifth, social welfare and inequality are functionals of the stationary distribution; welfare-improving interventions are characterized by their effect on the distribution via the Perron-Frobenius theorem.

The framework opens several directions. The endogenous multiplicity of equilibria in multi-agent extensions connects to the theory of institutional change. The continuous limit connects to classical incomplete-markets models. The monotone policy result suggests a theory of institutional poverty traps grounded in complementarity rather than non-convexity.

Most broadly, the personal economy framework suggests that the distribution of institutional identities — not just wealth or income — is a primary object of economic analysis and welfare evaluation. Improving economic welfare may require not merely redistributing resources within a fixed institutional structure, but changing the structure itself: expanding the number of attainable identities and improving the transition probabilities between them.

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# A Edge Count in Identity Lattices

**Proposition A.1.** *The number of edges in $G(\Omega)$ is*

$$E = \frac{N}{2}\sum_{k=1}^{n}(a_k - 1), \qquad N = \prod_{k=1}^{n} a_k.$$

*Proof.* By Theorem 3.2, every vertex has degree $\delta = \sum_k (a_k - 1)$. Summing over all $N$ vertices and dividing by 2 (each edge counted twice) yields $E = N\delta/2$. □

# B Proof of Theorem 3.5: Spectral Gap

The adjacency matrix of the Cartesian product of graphs satisfies the Kronecker sum formula:

$$A(G_1 \square \cdots \square G_n) = \bigoplus_{k=1}^{n} A(G_k) \;:=\; \sum_{k=1}^{n} I_{a_1} \otimes \cdots \otimes A(G_k) \otimes \cdots \otimes I_{a_n},$$

where $\otimes$ denotes the Kronecker product. The eigenvalues of a Kronecker sum are all sums $\sum_k \lambda_{s_k}(A(G_k))$.

The adjacency matrix of $K_m$ has eigenvalues $m-1$ (multiplicity 1, eigenvector $\mathbf{1}_m$) and $-1$ (multiplicity $m-1$, eigenvectors orthogonal to $\mathbf{1}_m$). Hence the eigenvalues of $A(G(\Omega))$ are all values of the form $\sum_k \mu_{s_k}$, where $\mu_{s_k} \in \{a_k - 1, -1\}$.

The largest eigenvalue is $\sum_k (a_k - 1) = \delta$, corresponding to $\mu_{s_k} = a_k - 1$ for all $k$ (the all-ones eigenvector).

The second-largest eigenvalue is obtained by changing exactly one factor from $(a_k - 1)$ to $-1$. The reduction in eigenvalue is $a_k$ for dimension $k$; this is minimized (second eigenvalue maximized) by choosing $k^* = \arg\min_k a_k$, giving:

$$\lambda_2(A) = \delta - \min_k a_k.$$

Since $G(\Omega)$ is $\delta$-regular and $P^{RW} = A/\delta$, we have $\lambda_2(P^{RW}) = (\delta - \min_k a_k)/\delta$ and the spectral gap $\gamma = 1 - \lambda_2(P^{RW}) = (\min_k a_k)/\delta$. □

# C Stationary Distribution of Birth-Death Chains

For a birth-death chain on $\{0, 1, \ldots, m-1\}$ with upward rate $\mu > 0$, downward rate $\nu > 0$, and reflecting boundaries, detailed balance yields:

$$\nu\,\pi(s+1) = \mu\,\pi(s) \quad \Longrightarrow \quad \pi(s) \propto \left(\frac{\mu}{\nu}\right)^s.$$

Normalizing over $s = 0, \ldots, m-1$:

$$\pi(s) = \frac{(\mu/\nu)^s}{\sum_{j=0}^{m-1}(\mu/\nu)^j} = \begin{cases} \dfrac{(1-r)\,r^s}{1-r^m} & \text{if } r = \mu/\nu \neq 1, \\ \dfrac{1}{m} & \text{if } r = 1, \end{cases}$$

where $r = \mu/\nu$. The mean state is:

$$\bar{z} = \sum_{s=0}^{m-1} s\,\pi(s) = \begin{cases} \dfrac{r}{1-r} - \dfrac{m\,r^m}{1-r^m} & r \neq 1, \\ \dfrac{m-1}{2} & r = 1. \end{cases}$$

For the example calibration in Section 8, with $(m, \mu, \nu) = (2, 0.10, 0.15)$ for the consumption dimension: $r = 2/3$, $\pi_4^*(0) = 3/5 = 0.60$, $\pi_4^*(1) = 2/5 = 0.40$, $\bar{z}_4^* = 0.40$, consistent with the numerical value cited in the text.

# The End